

1. Insurance Claim Prediction Report

Title

Insurance Claim Approval Prediction using Decision Tree

Objective

The objective of this case study is to build a machine learning model to predict whether an insurance claim will be approved or not. This helps in automating decision-making and improving efficiency in insurance processing.

Dataset

The dataset used is the Insurance Claims Dataset obtained from Kaggle. It contains various features such as customer details, vehicle information, and policy-related attributes that influence claim approval decisions.

Methodology

The dataset was first loaded and explored to understand its structure. Missing values were handled and unnecessary columns were removed. The data was then split into features and target variable.

A Decision Tree classifier was used to train the model. The dataset was divided into training and testing sets using train-test split. The model was trained on the training data and evaluated on the test data.

Result

The model achieved an accuracy of approximately **87%**, indicating strong performance in predicting insurance claim approvals. The Decision Tree algorithm successfully captured patterns in the dataset.

Conclusion

This case study demonstrates that machine learning models such as Decision Trees can effectively automate insurance claim prediction. With proper data preprocessing and model training, high accuracy can be achieved, making the system useful for real-world applications.